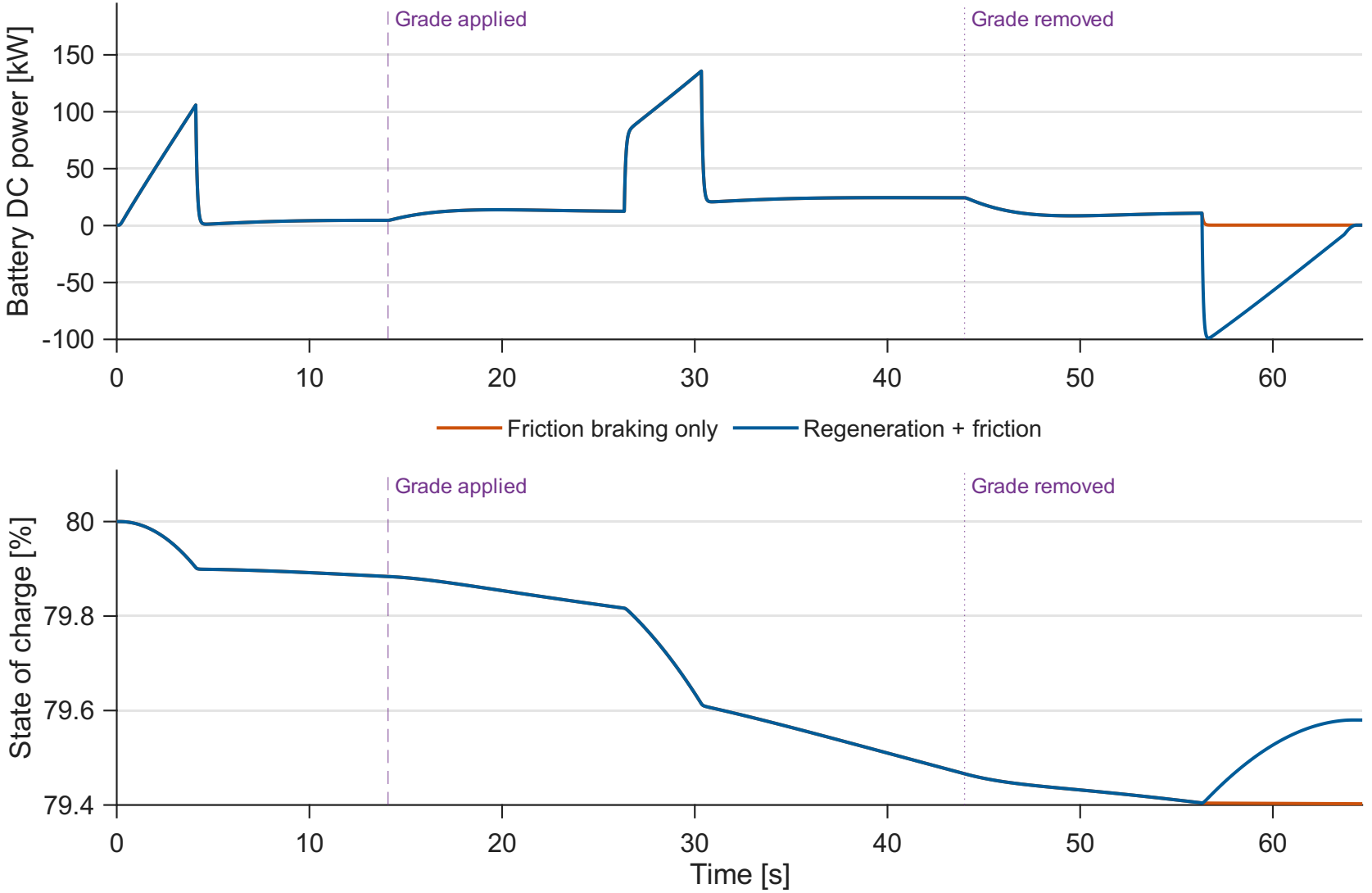

Grade applied
Grade removed
Battery DC power [kW]
150
100
50
0
-50
-100
0
10
20
30
40
50
60
Friction braking only
Regeneration + friction
Grade applied
Grade removed
State of charge [%]
80
79.8
79.6
79.4
0
10
20
30
40
50
60
Time [s]